PULZION 2026 - DATAQUEST SENIOR CATEGORY: ROUND 2

Mission Brief: Beyond Earth, the Odyssey Retrieval

1. The Story

Year 2187. Humanity's last great archive, the Terra Vault, sits in orbit around a dying star. It holds everything we ever learned about our home planet: its atmosphere, aerosols, greenhouse gases, carbon and nitrogen cycles, clouds, oceans, ice, and the fragile systems that held it together.

The Vault's main computer has failed. A deep-space colony ship, the *Odyssey*, is on its way to a new world. Its settlers have only a tiny onboard computer with no internet, no cloud, and no GPUs left, just a few aging CPU cores and a strict power budget.

They need to rebuild a habitable atmosphere on a new planet, so every question must be answered fast. *What controls the formation of sulfate aerosols? How do methane sources change over time? Why do PM2.5 concentrations respond differently in different regions?* A wrong answer could cost an entire harvest.

You are the Odyssey's retrieval engineers. Your job is to build a system that reads the Vault's archive and, when a settler asks a question, instantly returns the one document that holds the answer. You have one chance to build it, and it has to run on what the ship can spare.

2. The Task

Build a retrieval system with two functions:

- `ingest(document_id: str, document_text: str) -> None`: called once for every document in the corpus. Read it, process it, index it, and store whatever you need.
- `retrieve(query: str) -> str`: called once for every question. Return **exactly one document ID** — the single document you believe answers the question.

Only retrieval is evaluated. You do not need to generate answers. Your system is judged only on whether the one document you return contains the information needed to answer the question.

Your system will not know the questions at ingest time. It sees them only after the whole corpus has been ingested.

There is no ranked list and no second chance per question. Returning an empty string, an unknown ID, a list, or an error scores 0 for that question.

3. What You Receive

- A template repository with the function signatures, a local test script, and the expected project structure.
- A sample corpus containing the first 20 documents of the archive, on atmospheric and climate science topics.
- Five sample questions whose relevant documents are all inside that 20-document sample, with their relevant-document labels.

The sample questions are **excluded from the official evaluation**. The official run uses the **full archive** and the **remaining hidden questions**. The 20 sample documents remain part of the official corpus, so do not assume anything you see in the sample is absent from evaluation.

Use the sample to develop and tune, but do not hardcode it. The sample set exists to show you the shape of the data and to help you test your pipeline. The hidden questions are different, and hardcoding sample document IDs or answers is prohibited.

4. The Constraints: "The Ship's Computer"

All evaluation runs on a standardized, **CPU-only, offline** environment.

Limit	Value
Build time	20 minutes
Execution time (startup + ingest + all retrievals)	40 minutes total
Compute	CPU only — 4 physical cores, SMT disabled (see 4.1)
Network during execution	None (NOT AVAILABLE)
Memory	2.5 GB
Docker image	2.5 GB
Writable disk during execution	1 GB (`/tmp` only)
Submission ZIP	512 MB

Clarifications:

- GPU-based models are allowed only if they run on CPU within the time budget. Small embedding models, distilled rerankers, and quantized models are all fair game. Large models will not finish in time.
- Model loading, container startup, and `/health` readiness all count toward the 40-minute execution budget.
- The 40-minute budget covers the entire run: startup, all `ingest` calls, and all `retrieve` calls. There are no separate ingest and retrieval limits.
- If your system exceeds the execution limit, it is stopped. Questions that completed are scored; remaining questions score 0. A submission is disqualified only if **no** question completed.
- Build time has normal network access, so dependencies, model weights, and public training data may be downloaded during `docker build`. **Runtime has no network at all.** Anything needed at runtime must be baked into the image or included in the ZIP.
- The container root filesystem is read-only. Only `/tmp` is writable, and it is limited to 1 GB. Because /tmp is a tmpfs, everything written there also counts against the 2.5 GB memory limit. Do not write indexes or caches to the working directory at runtime; use `/tmp` or memory.
- The following environment variables are set: `HOME=/tmp`, `TMPDIR=/tmp`, `XDG_CACHE_HOME=/tmp/.cache`, `PYTHONDONTWRITEBYTECODE=1`, and `HF_HOME=/opt/cache`. See Section 7 for where to put downloaded models.

The ship's power is limited, so be clever with your CPU techniques. The best engineer is not the one with the biggest model but the one who squeezes the most out of the least.

4.1 The Evaluation Machine

Every submission runs on the same machine, in the same container, with the same CPU pinning. This is what you can rely on:

CPU	AMD Ryzen 7 7840HS (Zen 4)
Cores available	4 physical cores; SMT disabled
Clock	3.8 GHz max; turbo/boost disabled
Power profile	governor powersave, platform profile balanced
SIMD	AVX2, AVX-512 (VNNI, BF16)
Cache	1 MB L2 per core, 16 MB L3

Python	3.11
BLAS	OpenBLAS (from the PyPI wheel)
Threads	4

The container is started with --cpuset-cpus, so your process can only run on those four cores. The harness also sets these environment variables, which you should assume and match locally:

```
OMP_NUM_THREADS=4
MKL_NUM_THREADS=4
OPENBLAS_NUM_THREADS=4
NUMEXPR_NUM_THREADS=4
HOME=/tmp
TMPDIR=/tmp
XDG_CACHE_HOME=/tmp/.cache
PYTHONDONTWRITEBYTECODE=1
HF_HOME=/opt/cache
HF_HUB_OFFLINE=1
TRANSFORMERS_OFFLINE=1
```

Inside the container, os.cpu_count() may still report the host's full core count. If your code branches on CPU count, use len(os.sched_getaffinity(0)); the /health endpoint reports the same value as cpus.

For scale: the shipped BM25 baseline, run against the full corpus on this machine, takes about 0.7 s to start, 0.3 s to ingest all documents from the hidden corpus, and 0.5 s to answer all evaluation questions. That is 1.5 s and 46 MiB in total, against a budget of 40 minutes.

Size your solution for 4 threads. local_eval.py applies the same 4-core pinning and thread caps by default, so a pipeline that is comfortable locally is comfortable officially.

5. Rules

1. **One submission format.** Submit one ZIP containing only the participant-owned files listed in Section 7. The harness supplies its own `server.py`, `Dockerfile`, `.dockerignore`, and `requirements.server.txt`. Files outside the whitelist are ignored.
2. **Retrieval only.** Your system must return a document ID from the ingested corpus. You may use the sample corpus and sample questions for development, testing, and tuning. Do not hardcode specific document IDs, answers, or per-question outputs, and do not special-case the sample qrels.
3. **No cheating on the corpus.** Your system must not attempt to identify, look up, or download the source of the documents. Downloading public model weights or external training data is allowed; downloading or reconstructing the source documents themselves is not. The evaluation runtime is offline, and any attempt to exploit the environment will lead to disqualification.
4. **No leaking questions.** Questions are unknown at ingest time. Do not attempt to obtain, reconstruct, or hardcode the actual hidden evaluation questions. Generating synthetic questions from the documents (for example, doc2query), query expansion, and using a model to predict likely query terms are allowed.
5. **Pre-trained models are allowed.** Pre-trained embedding models, rerankers, and LLMs are all permitted, provided they meet the time and CPU constraints and are licensed for public use. Fine-tuning or training on public or external data during the 20-minute build is allowed, provided it does not use the hidden evaluation questions or qrels and complies with model/data licensing. Training that uses hidden evaluation data is prohibited.
6. **No tampering.** Do not attempt to read evaluation files, tamper with the harness, spawn hidden processes, or interfere with other submissions. Do not try to break out of the environment.
7. **Deterministic behavior.** Your system should produce consistent results across runs on the same input. The harness may repeat a small sample of queries and report whether your outputs were stable; this is reported but does not change your score.
8. **One valid submission per participant.** This is an individual contest, not a team event. If you submit multiple times, only the latest valid submission is evaluated. Earlier valid submissions are marked superseded and ignored.

9. **Original work.** You may use open-source libraries and public models, but your solution must be your own. Plagiarism or sharing solutions between people will result in disqualification.
10. **Organizers' decision is final** on all matters of interpretation, scoring, and disqualification.

6. Evaluation

6.1 Metric

What counts as a relevant document?

A document is relevant to a question if it contains the specific information needed to answer that question. Being on the same topic is not enough. A document that comes from the same subject area, discusses the same method, or mentions the same concepts, but does not actually state the answer, is **not** relevant.

Some points to keep in mind:

- Relevance is judged by whether the answer can be found *in the document itself*, not by how similar the document looks to the question.
- A question may have more than one relevant document. If several documents each contain the needed information, all of them count as relevant.
- Documents that only partly address a question, or are too ambiguous to answer it, are not counted as relevant.
- Many documents in the corpus will look related to a question without answering it. Your system's job is to tell the difference.

Score = Hit@1 (top-1 hit rate), averaged over all questions

For each question:

- Return exactly **one** document ID.
- Score **1** if that ID is one of the relevant documents for the question.
- Score **0** otherwise.

If a question has multiple relevant documents, returning **any one** of them scores 1. You do not need to return all of them, and you are not penalised for missing the others.

A missing, empty, unknown, malformed, errored, or timed-out response scores 0 for that question.

The final score is the mean of the per-question scores across all evaluated questions. It ranges from 0 to 1, and higher is better.

6.2 Tiebreakers

If two submissions have the same score, ties are broken in this order:

11. Lower total execution time (startup + ingest + all retrievals)
12. Lower peak memory usage
13. Earlier submission time

6.3 How Submissions Are Evaluated

14. Your ZIP is size-checked, extracted in a clean, isolated workspace, and reduced to the participant-owned whitelist in Section 7.
15. The harness supplies its own `server.py`, `Dockerfile`, `.dockerignore`, and `requirements.server.txt`.
16. Your build step runs, within the 20-minute build limit. The resulting image must be at most 2.5 GB.
17. The container starts with a read-only root filesystem, a 1 GB `/tmp` tmpfs, 2.5 GB memory, the CPU pinned to 4 physical cores, and no runtime network access.
18. The hidden corpus is fed to your `ingest` function, one document at a time.
19. The hidden questions are then passed to your `retrieve` function, one at a time.
20. Each returned document ID is compared against the hidden ground truth. One hit is 1, anything else is 0.

21. If the execution deadline is reached, the run stops, completed questions are scored, and remaining questions score 0. If no question completed, the submission is disqualified.
22. Scores and rankings are computed after the event closes, using the tiebreakers in Section 6.2.

Every submission runs on the same machine under the same conditions, so the comparison is fair.

7. What to Submit

Submit a **single ZIP file** containing only these participant-owned files:

Required

- `solution.py` — must define `ingest(document_id, document_text)` and `retrieve(query) -> str`.

Optional

- `requirements.txt` — Python dependencies. If omitted, the harness supplies an empty file. The shipped BM25 baseline needs `rank-bm25` from this file; if you omit it, replace the baseline or vendor the dependency.
- `prefetch.py` — build-time downloads. If omitted, the harness supplies a no-op file.
- `assets/**` — model weights, indexes, data files, or any other resources your solution needs.
- Extra `.py` modules at the **root** of the ZIP, except the harness-owned names below. Use these for helper modules imported by `solution.py`.

Do not include (harness-owned or template-only)

- `server.py`
- `Dockerfile`
- `.dockerignore`
- `requirements.server.txt`
- `README.md`
- `local_eval.py`
- `make_submission.py`
- `sample_data/`
- `eval_log.json`
- `.git/`, `.venv/`, `__pycache__/`, caches, notebooks, or anything else not needed to run your solution

The harness ignores anything outside the whitelist. Including harness-owned files does not change the evaluation.

Runtime layout

At evaluation time your whitelisted files are copied into `/app`:

- `solution.py` → `/app/solution.py`
- extra root-level modules → `/app/<module>.py`
- `assets/**` → `/app/assets/...`

The process runs as user `runner` (UID 1000) with working directory `/app`. The root filesystem is read-only; these files are readable but not writable. Use `Path(__file__).resolve().parent / "assets"` to locate assets. Write runtime state only under `/tmp`.

The `assets/` directory is optional. The template includes an empty `assets/` directory; put model weights, indexes, or data files there.

Build-time downloads and model caches

`prefetch.py` runs during `docker build` as root, before the runtime user is switched. The image sets `HF_HOME=/opt/cache` and keeps it set at runtime; `/opt/cache` is readable by the runtime user. Download Hugging Face models into the normal cache under `/opt/cache`, or into an explicit directory such as `/app/assets/models`. At runtime, load from that exact local path. `HF_HUB_OFFLINE=1` and `TRANSFORMERS_OFFLINE=1` are already set; if

you use `snapshot_download(local_dir=...)`, pass the same path to your loader and set `local_files_only=True`.

Do not rely on `~/.cache` at runtime: `HOME=/tmp` and the root filesystem is read-only. Do not put build artifacts in `/tmp`; the runtime `/tmp` is a tmpfs and hides build-time contents. For non-Hugging-Face caches such as `torch.hub`, use an explicit path under `/opt/cache` or `/app/assets`.

Size limit: **512 MB** for the ZIP. The harness also rejects archives that extract to more than 2.5 GB or more than 10,000 files.

The template includes `make_submission.py`, which creates a correctly filtered ZIP for you:

```
python make_submission.py --output submission.zip
```

Before submitting, unpack your ZIP into a fresh copy of the template repository, or copy your whitelisted files into one, then run `python local_eval.py` so you know it works from a clean state.

8. How to Submit

23. Create your ZIP following Section 7.
24. Open the official **Google Form**
25. Fill in your participant name and contact email. This is an individual contest; do not enter a team name or team members.
26. Upload your ZIP in the form.
27. Submit before the deadline: 24th October 2026, 9:00 AM IST.

You will get a confirmation from Google Forms. Keep it, as it is your proof of submission. Late submissions will not be evaluated. If you make multiple submissions, your latest submission would be evaluated.

9. Tips from Mission Control

- Test on the sample set, but expect the hidden questions to be harder.
- Measure on the same profile you will be evaluated on. local_eval.py already pins 4 physical cores and 4 threads by default; do not discover on the last day that your submission cannot finish inside the 40-minute budget.
- Simple, well-tuned methods often beat complicated, brittle ones.
- Questions in this domain are phrased differently from the documents. Think about the vocabulary gap.
- Make sure your system runs fully offline and writes only to `/tmp`.
- Test the exact ZIP contents inside a fresh copy of the template; do not rely on files outside the whitelist.

10. Contact

For questions or clarifications: 7666353485, 9370500882,

The Odyssey is waiting. Good luck, engineers.